Spillovers from workers' compensation to Social Security Disability Insurance: evidence from Oregon

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Abstract

The two most important safety nets for workers suffering occupational injuries and illnesses are workers' compensation programs and Social Security Disability Insurance. Workers' compensation is the oldest social insurance program in the United States and Social Security Disability Insurance is the largest provider of cash benefits to individuals with a disability. The programs are linked because the injuries and illnesses that workers suffer can satisfy the requirements of both programs. In this paper, I examined the impact of an Oregon law (SB 757) that changed the calculation of a type of benefit provided under workers' compensation called permanent partial disability. This study was designed to answer whether these changes led to spillover effects to applications for Social Security Disability Insurance. Using a difference-in-differences framework applied to data from the Social Security Administration, I did not find statistically significant results indicative of such effects. (JEL No. I18, J28, J32)

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1 Introduction

The two most important sources of benefits for individuals suffering occupational OIIs and illnesses (OIIs) are workers' compensation programs and Social Security Disability Insurance. Workers' compensation (WC) is the oldest form of social insurance in the United States, dating back to the early 20th century. Social Security Disability Insurance (DI) is the largest provider of cash benefits in the United States to individuals with a disability. WC is primarily administered by each state for its residents, while DI is a federal program.

WC and DI are linked because the OIIs that individuals suffer can result in circumstances that satisfy the requirements of both programs. Individuals with OIIs whose resulting impairment is expected to last for 12 months or result in death may be eligible for both WC and DI. Since the policy parameters of WC programs change often (Qiu and Grabell, 2015), this could presumably affect the rate of DI applications at the margin. Hence, in this paper, I seek to answer the question: How do changes in workers' compensation law spillover to applications for benefits from Social Security Disability Insurance?

To answer this question, I examined the impact of an Oregon law passed in 2003. In 2003, Oregon enacted SB 757, which amended the calculation of a type of WC called permanent partial disability (PPD). The changes in SB 757 applied to OIIs suffered on or after January 1, 2005. The changes meant that some individuals awarded PPD who were deemed unable to return to work received higher WC cash benefits.

The changes from SB 757 served as plausibly exogenous variation to identify how individuals respond to changes in WC benefit amounts via their DI application behavior. I tested this empirically using workload data from the Social Security Administration and a difference-in-differences approach. I defined three separate control groups: the state of Washington, a control with empirically derived weights, and an equal weight control. I estimated the treatment effect against each separately. All of my estimates implied treatment effects that I could not statistically distinguish from zero. In other words, I found no spillover effect from this change in Oregon's WC law to DI applications.

My paper contributes to the growing literature on the interactions between the two programs, and is to my knowledge the first paper to study these interactions using a difference-in-differences approach. There are two main themes in the literature. One theme is that costs of OIIs, which should have been borne by WC programs, are instead being paid through DI. Reville and Schoeni (2004) showed using data from the Health and Retirement Study in 1992, for instance, that only 12 percent of respondents aged 51 to 61 with OIIs received WC benefits, but 29 percent had received DI. Leigh (2011) estimated that cost shifting along these lines means that WC covered less than 25 percent of the $250 billion of total costs of OIIs in 2007. This cost shifting may be especially prominent among more severe OIIs, as O'Leary et al. (2012) found in New Mexico.

The other theme in the literature, which is more directly addressed by my paper, involves the effects of WC policies. Prior work has examined the effects of WC programs becoming more restrictive, starting in the 1980s. These more restrictive policies were enacted because of employer concerns about the increasing costs of providing WC benefits. Authors have hypothesized that individuals suffering OIIs substitute away from WC towards DI as WC becomes relatively less attractive. There is mixed evidence on this point empirically, however. Whereas Guo and Burton (2012) and Buffie and Baker (2015) found evidence of this effect, McInerney and Simon (2012) do not.

My paper proceeds as follows. Section 2 provides the background of the two programs. Section 3 motivates the empirical approach. Section 4 describes the data, control groups, and offers summary statistics. Section 5 discusses the results. Section 6 concludes.

2 Background

2.1 Workers' Compensation

Before states established WC programs, individuals had two main avenues of seeking compensation for OIIs (Fishback and Kantor, 2000).[1] Prior to an OII, individuals needed to negotiate higher wages to compensate for more dangerous jobs; individuals could purchase individual workplace accident insurance policies by paying premiums with these wage differentials.[2] After an OII, individuals needed to reach a settlement with their employers or file suit against them for negligence. That is, individuals could recoup their financial losses along with compensation for pain and suffering through the court system. In practice, litigation resembled a lottery: injured individuals often received nothing because of employer defenses, but in a handful of cases recovered large sums. This type of system eventually proved untenable. WC represented a shift to a system based on no-fault liability that paid regardless of whether the employer or employee was responsible for an OII.

WC benefits comprise both medical and indemnity (*i.e.,* cash) payments. The majority of approved WC claims are resolved as medical only (Welch et al., 2024). The amount of indemnity payments an injured individual receives depends on OII duration and extent. Temporary total disability covers an injured individual who cannot immediately return to work. Temporary partial disability applies to individuals who return to work, but at reduced hours or wages. Impairments that remain after reaching so-called *maximum medical improvement* qualify for permanent benefits. At maximum medical improvement, the impairment is stable and not expected to get better with additional treatment. Insurers pay for permanent total or permanent partial disability based on the severity of an OII. Individuals can receive an income from these WC benefits for the rest of their

[1] Before states established WC programs, I am not aware of no-fault insurance policies that would have compensated individuals for on-the-job OIIs through medical and cash payments. Employers held liability insurance that insured them against lawsuits; employers also reached settlements with employees that compensated them for lost wages and medical expenses. But this system was not no-fault. Thus, if employers believed the employee had any fault for the OII, they often paid nothing.

[2] These policies were not no-fault.

lives. In 2019, the value of all WC benefits paid across all states was $63 billion.[3]

2.2 Social Security Disability Insurance

During the Great Depression, FDR signed into law the Social Security Act of 1935 as part of his New Deal to support the economic security of Americans. The Act included benefits for retired individuals but was silent on benefits for individuals with disabilities. Discussions to add these benefits began in 1936, but there was debate over how stringent to make the definition of disability. It took 20 years for Congress to pass the Social Security Act Amendments of 1956 and include benefits for individuals with disabilities. The Social Security Administration (SSA) administers the DI program, which is funded through payroll taxes on American workers.

Individuals with disabilities must meet two requirements to receive DI benefits. First, individuals must financially qualify for DI by earning work credits. Individuals earn up to four work credits each year by earning a certain amount of income. For 2022, individuals earn one work credit for each $1,510 of income. Social Security rules require applicants to have an age-dependent number of work credits.[4] Second, individuals must medically qualify for DI by having a disability that meets the Social Security definition. The definition of disability used by DI is strict. Individuals must have a disability that is expected to last for at least 12 months or result in death, while being unable to engage in substantial gainful activity. An individual is determined to be unable to engage in substantial gainful activity based on the available medical and vocational evidence. A federally funded state agency known as a Disability Determination Service makes this decision once the individual is found to financially qualify. If the Disability Determination Service awards benefits, the beneficiary receives them until retirement subject to continuing disability review. In 2019, the SSA paid DI benefits of $126 billion to individuals with disabilities.[5]

[3]The WC program is more generous in terms of access to benefits than DI.

[4]For example, individuals who apply for DI with a disability that started between ages 31 and 42 need 20 credits.

[5]This makes the DI program significantly more generous in terms of total spending than WC.

2.3 Cost Shifting from Workers' Compensation to Social Security Disability Insurance

WC and DI are linked because OIIs occasionally result in circumstances that satisfy the requirements of both programs. Individuals receiving benefits from both programs have their total amount offset (*i.e.,* capped) at 80% of their average current earnings prior to disability. As Guo and Burton (2016) observed in a report for the Committee for a Responsible Federal Budget, not all instances in which individuals receive benefits from both programs involve cost *shifting*. Assuming adequate benefits provided by WC, costs may instead be *shared* by both programs up to the level of the offset. However, the adequacy standard for WC benefits—replacement of two-thirds of an individual's lost wages—is often not met. State-level policies may provide inadequate WC benefits through the restriction of the amount of benefits paid, the duration for which they are paid, or the eligibility requirements for receiving them. In these instances, costs are shifted onto the DI program when individuals receive benefits under both programs. This creates tension between the two programs. Whereas WC is financed by experience-rated employer premiums, DI is funded by federal payroll taxes on almost all employees. Consequently, inadequate WC benefits effectively transfer financial liability from employers to federal taxpayers.

While the offset represents the formal relationship between the two programs, a 2023 report of the SSA's inspector general highlighted two key problems with its application (Office of the Inspector General, Social Security Administration, 2023). First, the SSA lacks data matching agreements with states. Even though the SSA asks about WC receipt when individuals apply for DI benefits, this disclosure is a self-report. This makes the implementation of the offset prone to a substantial degree of error. Second, the SSA must honor the reverse offset plans of 14 states.[6] Unlike regular offset, in which DI benefits are reduced if combined benefits exceed the 80% threshold, reverse offset plans reduce WC benefits instead. This results in payments that the SSA could have otherwise not made had the regular offset provisions been in place. In its 2018 budget,

[6]The reverse offset states are Alaska, California, Colorado, Florida, Louisiana, Minnesota, Montana, New Jersey, New York, North Dakota, Ohio, Oregon, Washington, and Wisconsin. Federal law bans the establishment of new offset plans.

the first Trump administration proposed eliminating all reverse offset plans, but that proposal did not become law.

In the literature, several studies provide evidence of potential cost shifting. For instance, in one of the first studies on this topic, Reville and Schoeni (2004) found that among older individuals with work-related health conditions, 29 percent participated in DI while only 12 percent had ever received WC. Additionally, Leigh and Marcin (2012) estimated that of the $183 billion in non-medical costs of OIIs in 2007, only $22 billion was paid for by WC programs. In New Mexico, O'Leary et al. (2012) showed that individuals with lost-time impairments were significantly more likely to receive DI benefits than those who received medical only benefits. This occurred because the lost wages from these impairments persisted or intensified after the duration of WC receipt, indicating inadequate WC benefit payments.

Finally, McInerney and Simon (2012) reached a different conclusion as did Guo and Burton (2012) and Buffie and Baker (2015) on whether increasingly restrictive WC policies shift costs to DI. The conflicting findings in the literature arise from differences in how the authors measured program restrictions. McInerney and Simon utilized state maximum weekly benefits as their primary policy variable and found no statistically significant evidence of substitution using an instrumental variables approach. However, Guo and Burton argue that maximum benefits are a limited measure that fails to capture the full scope of restrictiveness. When employing broader measures like actuarial expected benefits, they found a significant relationship between WC tightening and DI applications in a fixed effects framework. Buffie and Baker found similar evidence of spillovers using beneficiary caseloads. This debate highlights the potential value of analyzing a discrete legislative shock.

3 Motivation and SB 757

To explore the linkages between WC and DI empirically, I exploited plausibly exogenous variation as a result of a law that changed the calculation of permanent partial disability (PPD) benefits in Oregon.[7] Known as Senate Bill 757 (SB 757), this law allowed the *social-vocational factor*, an input into the calculation of PPD benefits, to apply to all OIIs on or after the implementation date of January 1, 2005. Previously, certain OIIs were ineligible for inclusion of the social-vocational factor, even if those OIIs reduced an individual's ability to return to work. These OIIs were known as *scheduled* OIIs. There were approximately 15 scheduled OIIs, and reflected OIIs that affected the bodily extremities, vision, or hearing. SB 757 eliminated the distinction between scheduled and unscheduled OIIs.[8]

The impetus for SB 757 was the 2001 Oregon Supreme Court decision in *Smothers v. Gresham Transfer, Inc.* (Department of Consumer and Business Services, 2006). That decision permitted individuals whose WC claim was denied to sue their employer for OII-related costs if the reason for the denial was that their OII was not the "major contributing cause" of their disability. The court's reasoning was that such denials failed to provide these individuals with an adequate remedy under Oregon's Constitution. Against this backdrop, and afraid of increased costs to employers, the legislature mandated a proposal for its 2003 session. The Management-Labor Advisory Committee—originally established in 1990 to help stabilize costs of the Oregon WC system—fulfilled this mandate by presenting a proposal that enhanced benefit adequacy by better matching compensation to lost earnings. Importantly, in order to garner the support of employers, the proposal sought to keep costs flat. To achieve this objective, the proposal meant that individuals who could not return to work would be paid more whereas individuals who could return to work would be paid less. This proposal became SB 757.

Specifically, the law made changes as can be seen in the formulas below, where PPD_{it} is the

[7] In Oregon, PPD benefits are paid monthly by a WC insurer up to the calculated PPD benefit amount with two exceptions. One exception is that any PPD award not exceeding $6,000 must be paid in a lump sum. The other exception is that an individual may receive approval from the WC insurer for a lump sum payment. The offset of a WC lump sum payment on DI benefits is calculated by SSA *as if* the WC payments were made monthly.

[8] Oregon Administrative Rule 436-035-0012 states that for an OII prior to January 1, 2005, an individual must have an unscheduled impairment to have a calculated social-vocational factor.

PPD benefit amount of individual i suffering an OII in year t:

$$\text{PPD}_{it} = \begin{cases} p_i^m \cdot D^S \cdot B_t^S & \text{if } S = 1 \\ f((p_i^m + V_i \cdot p_i^v) \cdot 320) & \text{if } S = 0 \end{cases} \qquad \text{(Prior to 1/1/2005)}$$

$$\text{PPD}_{it} = p_i^m \cdot 100 \cdot SAWW_t + V_i \cdot (p_i^m + p_i^v) \cdot 150 \cdot w_{it} \qquad \text{(On or After 1/1/2005)}$$

For OIIs prior to January 1, 2005, the calculation included an impairment percent p_i^m of the maximum number of degrees based on the type of OII. For scheduled OIIs ($S = 1$) the maximum number of degrees, D^S, was specified body part by body part, including hearing and vision loss. As an example, the loss of function or use of an arm at or above the elbow joint had a maximum of 192 degrees. There was a fixed dollar amount awarded, B_t^S, per scheduled degree, regardless of body part, including hearing and vision loss. If there were multiple scheduled OIIs, each would be determined and awarded separately. For unscheduled OIIs ($S = 0$), the maximum number of degrees was 320 degrees and the impairment percent was considered for the whole person. If the individual was not released to regular work, then they were eligible ($V_i = 1$) for inclusion of the social-vocational factor p_i^v. If the individual did return or was released to regular work, then they were not eligible ($V_i = 0$) for inclusion of the social-vocational factor. Finally, there was a convex kinked function, $f(\cdot)$, that was increasing in the degrees of unscheduled impairment that determined the benefit amount.[9]

For OIIs on or after January 1, 2005, the impairment percent p_i^m is expressed as a percent of the whole person. The impairment percent is multiplied by 100 times Oregon's state average weekly wage at the time of the OII, $SAWW_t$, to arrive at the impairment benefit. If the individual is not released to regular work, then they are eligible ($V_i = 1$) for inclusion of the social-vocational factor p_i^v as part of a term added to the impairment benefit. This second term is known as the work disability benefit. To calculate the work disability benefit, the social-vocational factor p_i^v is added to the impairment percent p_i^m. This sum is multiplied by 150 times the individual's average weekly

[9]The convex kinked function in effect on December 31, 2004, was \$184 per degree for degrees 0-64, \$321 per degree for degrees 64.1-160, and \$748 per degree for degrees 160.1-320.

wage w_{it}, which is bound between 50 and 133 percent of $SAWW_t$, inclusive.

Inclusion of the social-vocational factor in the PPD benefit is contingent on the individual not being released to regular work. This determination is made by an individual's attending physician once the individual is determined to reach maximum medical improvement. Individuals receiving WC benefits in Oregon must choose an attending physician qualified under Oregon law. Individuals who are cleared by their attending physician to return to their regular job do not have a social-vocational factor. On the other hand, the work disability benefit is not reduced once awarded. In other words, there is no restriction on later returning to work if work disability has been awarded.

The social-vocational factor p^v is calculated by the following formula:

$$p^v = \frac{(\text{Age Factor} + \text{Education Factor}) \times \text{Adaptability Factor}}{100} \times 100\%$$

The Age Factor is 1 for individuals aged 40 and over and 0 otherwise. The Education Factor ranges from 0 to 5, and is the sum of values based on an individual's formal education and specific vocational preparation. For the formal education value, individuals get a value of 1 if they do not have a high school diploma or GED and 0 otherwise. For the specific vocational preparation value, individuals get a value from 1 to 4 based on the training time needed to perform their jobs in the past five years. They receive a value of 4 if needing only a short demonstration, a value of 3 if needing 30 days to 6 months, a value of 2 if needing 6 months to 2 years, and a value of 1 if needing more than 2 years. Finally, the Adaptability Factor ranges from 1 to 7 and is based on an individual's impairment rating or a comparison of the individual's base functional capacity (prior to their OII) to their residual functional capacity (at maximum medical improvement), whichever results in a higher Adaptability Factor.[10] Individuals with the most severe impairments receive the maximum value of 7; values decrease to 1 as impairments become less severe.

Consider the stylized example of the OIIs shown in Table 1 for two time periods: prior to January 1, 2005, and on or after that date. There are four scenarios needed to explore the differences in

[10] Functional capacity refers to the weight of objects an individual can lift or carry; this ranges from sedentary restricted (less than 10 pounds) to very heavy (occasionally lifting over 100 pounds and frequently lifting or carrying more than 50 pounds).

Table 1: SB 757 Example

Panel A: Prior to 1/1/2005

	Arm Injury RTW	Arm Injury	Back Injury RTW	Back Injury
Scheduled	Y	Y	N	N
Return to Work	Y	N	Y	N
Maximum Degrees	192	192	320	320
Impairment Percent	30%	30%	18%	18%
Degrees of Impairment	57.6	57.6	57.6	57.6
Social-Vocational Factor	N/A^1	N/A^1	N/A^2	6%
WC Benefit	$32,198	$32,198	$10,598	$14,131
Pre-OII Annual Labor Income	$46,800	$46,800	$46,800	$46,800
Annual DI Benefit	N/A^2	$24,000	N/A^2	$24,000
Year 1 WC Benefit	$31,320	$31,320	$10,598	$14,131
Year 2 WC Benefit	$878	$878	$0	$0
Maximum Yearly WC+DI Income	$37,440	$37,440	$37,440	$37,440
Year 1 WC+DI Income	$31,320	$37,440	$10,598	$37,440
Year 2 WC+DI Income	$878	$24,878	$0	$24,000
First 2 Years WC+DI Income	**$32,198**	**$62,318**	**$10,598**	**$61,440**

Panel B: On or After 1/1/2005

	Arm Injury RTW	Arm Injury	Back Injury RTW	Back Injury
Return to Work	Y	N	Y	N
Impairment Percent	18%	18%	18%	18%
Social-Vocational Factor	N/A^2	6%	N/A^2	6%
WC Benefit	$12,394	$44,794	$12,394	$44,794
Pre-OII Annual Labor Income	$46,800	$46,800	$46,800	$46,800
Annual DI Benefit	N/A^2	$24,000	N/A^2	$24,000
Year 1 WC Benefit	$12,394	$31,320	$12,394	$31,320
Year 2 WC Benefit	0	$13,474	0	$13,474
Maximum Yearly WC+DI Income	$37,440	$37,440	$37,440	$37,440
Year 1 WC+DI Income	$12,394	$37,440	$12,394	$37,440
Year 2 WC+DI Income	$0	$37,440	$0	$37,440
First 2 Years WC+DI Income	**$12,394**	**$74,880**	**$12,394**	**$74,880**

RTW: Return to Work. N/A: Not Applicable. [1]Not Applicable Due to Scheduled OII. [2]Not Applicable Due to Return to Work. Based on end of 2004 values for Scheduled OIIs ($559.00 per degree) and Unscheduled OIIs ($184.00 per degree). Based on beginning of 2005 value for State Average Weekly Wage ($688.56 per week). Year 1 WC Benefit is lesser of WC Benefit or calculation according to Oregon Administrative Rule 436-060-0040 of 12 times 4.35 times 66 2/3 % of individual weekly wage of $900. Year 2 WC Benefit is WC Benefit minus Year 1 WC Benefit. Maximum Yearly WC+DI Income calculated by applying offset of 80% to Pre-OII Annual Labor Income. Year 1 WC+DI Income is the lesser of Annual DI Benefit plus Year 1 WC Benefit or Maximum Yearly WC+DI Income. Year 2 WC+DI Income is the lesser of Annual DI Benefit plus Year 2 WC Benefit or Maximum Yearly WC+DI Income.

the PPD calculations within and between each time period. I first considered an arm injury with and without the individual being cleared to return to work. I next considered a back injury with and without the individual being cleared to return to work. The individual has a pre-OII annual labor income of $46,800. Based on Oregon's WC law, this means that the annual WC benefit can be a maximum of $31,320.[11] If the individual does not return to work, I considered that individual receiving an annual DI benefit of $24,000 to show how any DI benefit interacts with the offset. Combined, this meant that the maximum WC and DI income in the first year can be $37,440 (= 80% × $46,800).[12] In these scenarios, I assumed that the WC benefit and any DI benefit start in January in Year 1.

Panel A shows examples of the calculation of PPD benefits for OIIs suffered prior to January 1, 2005. First, an arm injury was a scheduled OII. The example shows that this scheduled OII had a maximum of 192 degrees. Assuming an impairment percent of 30%, the degrees of this scheduled impairment would be 57.6. Because this was a scheduled OII, it was not eligible for inclusion of the social-vocational factor in the calculation of the PPD benefit, regardless of whether the individual was cleared to return to work. At the end of 2004 value for scheduled OIIs of $559.00 per degree, this translated into a WC benefit of $32,198, of which $31,320 was paid in the first year and $878 paid in the second year. If the individual did not return to work, the individual would have a first year DI benefit of $24,000, but because of the offset would receive $37,440 in combined WC and DI benefits in the first year and $24,878 in the second year for a total of $62,318. Second, a back injury was an unscheduled OII. The example shows that this unscheduled OII was considered as a percent of the whole person, so had a maximum of 320 degrees. Assuming an impairment percent of 18%, the degrees of this unscheduled impairment would be 57.6. If the individual was not cleared to return to work, the example assumes a realistic social-vocational factor of 6%. At the end of 2004 value for unscheduled OIIs of $184.00 per degree for the first 64 degrees, this translated into a WC benefit of $10,598. This full amount would be paid to the individual if they returned to

[11] This is calculated based on Oregon Administrative Rule 436-060-0040 by multiplying the individual's weekly wage by 12 times 4.35 times 66 2/3 %.

[12] Since Oregon is a reverse offset state, the full $24,000 DI benefit would be paid and the WC amounts in excess of the offset would not be paid.

Table 2: Estimated Group Sizes of Permanent Partial Disability Awards, Oregon, 2001-2004

Panel A: Estimated Yearly Counts, 2001-2004

	Scheduled	Unscheduled	Total
Return to Work	2,900	2,000	4,900
No Return to Work	1,000	600	1,600
Total	3,900	2,600	6,500

Panel B: Estimated Yearly Percentages, 2001-2004

	Scheduled	Unscheduled	Total
Return to Work	45%	31%	75%
No Return to Work	15%	9%	25%
Total	60%	40%	100%

Sums may not equal totals due to rounding.
Estimates based on Department of Consumer and Business Services (2006) and Mullen and Rennane (2024).

work in the first year. On the other hand, if the individual was not cleared to work, the individual would receive $37,440 in combined WC and DI benefits in the first year because of the offset and $24,000 in the second year for a total of $61,440.

Panel B shows examples of the calculation of PPD benefits for OIIs suffered on or after January 1, 2005. For these OIIs, there is no consideration of scheduled versus unscheduled nor degrees of impairment. All impairments are rated as part of the whole person. Furthermore, the social-vocational factor applies to any individual not cleared to return to work. The amounts paid to the individual who returned to work for an arm injury or back injury were thus the same. With the values assumed in this example, that amount is $12,394 in the first year and $0 in the second year. Likewise, the amounts paid to the individual who was not cleared to return to work for an arm injury or back injury were also the same. With the values assumed in this example, that amount is $37,440 in each of the first two years because of the offset, for totals of $74,880.

The substantial difference in benefit amounts between the scenarios discussed above highlighted the importance of understanding how frequently each occurred. That is, to determine the potential aggregate impact of SB 757 on applications for DI benefits, I needed to estimate the sizes of these groups. Based on estimates contained in Department of Consumer and Business Services

(2006) and Mullen and Rennane (2024), along with aggregated PPD claim counts from Oregon, I estimated yearly counts and percentages along two dimensions: whether an OII was scheduled or unscheduled and whether the individual was cleared to return to work or not.

Table 2 shows estimates for the years 2001 through 2004. As can be seen in the table, of the 6,500 PPD awards each year, 60 percent were due to scheduled OIIs and 40 percent were due to unscheduled OIIs. Conversely, in 75 percent of awards the individual returned to work whereas in 25 percent of awards the individual was not cleared to return to work. Spillover effects to applications for DI benefits would occur in only those cases where the individual was not cleared to return to work. This is because returning to work implies engaging in substantial gainful activity, which would disqualify an individual for the receipt of DI benefits. Presumably these individuals would therefore not attempt applying for DI benefits.

The implementation of SB 757 significantly altered the landscape for the estimated 45 percent of individuals who would have had scheduled OIIs under the prior structure. As shown in Table 1, individuals with scheduled OIIs received the same PPD benefit regardless of their return-to-work status, creating no financial incentive within the prior structure to exit the labor force. For these individuals, returning to work effectively now means accepting a significantly lower benefit than they would have before. On the other hand, not being cleared to return to work opens access to the new, higher work disability awards. This provides a significant incentive to try and not be cleared to return to work. For individuals who would have had unscheduled OIIs under the prior structure, the incentives changed less strongly. These individuals already had the possibility of inclusion of the social-vocational factor within their PPD award. However, despite these weaker changes, the law may nonetheless have strengthened the incentive for some of these individuals to remain out of the labor force, as well.

Based on this analysis, my main hypothesis centered on the responsiveness of individuals, particularly those with previously scheduled OIIs, to these changed incentives. I hypothesized that these changes would lower the opportunity cost of exiting the labor force, leading to an overall increase in DI applications. This would be because of the increased support to these individuals while remaining out of the labor force and not engaging in substantial gainful activity. Individuals would then use this period in which they were not engaging in substantial gainful activity to apply for DI benefits. To evaluate this predicted increase, I used aggregated data on applications from the SSA.

4 Data, Control Groups, and Summary Statistics

4.1 Data

The data on DI applications comes from the State Agency Monthly Workload file published by the SSA. DI applications can be submitted as DI only applications, or concurrently with applications for Supplemental Security Income, a federal program for applicants with limited resources and limited work history. Because I am interested in potential spillovers that occurred due to changes in WC law and want to consider individuals with work history, I restrict what follows to applications for DI only. The State Agency Monthly Workload file shows the DI applications referred to each state's Disability Determination Service by year and month. The SSA typically forwards an application to the Disability Determination Service of the applicant's state of residence. A limitation of this dataset is that workload data do not perfectly correspond to residence data due to SSA processing idiosyncrasies. While I do not have access to data based on applicant residence, I expect my data closely matches this ideal. This is because the primary purpose of a state Disability Determination Service is to review applications from residents of that state.

The response variable in my equations was a state's number of DI only applications divided by that state's population of adults aged 25-64. This rate is multiplied by 100,000 for ease of exposition. The numerator had monthly observations of DI only applications from the State Agency Monthly Workload file. The denominator had yearly observations of population from the Expanded State Employment Status Demographic Data. The Expanded State Employment Status Demographic Data is a product of the Bureau of Labor Statistics and includes state-level data on population by year. The response variable was calculated for each state at the year-month level. As covariates, I

Table 3: State-Level Weights for the Control with Empirically Derived Weights

State	Weight
Arizona	0.031
Colorado	0.102
Hawaii	0.197
Idaho	0.116
New Mexico	0.118
Utah	0.189
Washington	0.220
Wyoming	0.027

Weights are calculated empirically based on the method developed by Arkhangelsky et al. (2021).

included the unemployment rate and the female employment share. I calculated these rates from the basic monthly Current Population Survey obtained via IPUMS (Flood et al., 2024).[13] Like in the response variable, these covariates were restricted to individuals aged 25-64. These covariates were calculated for each state at the year-month level, as well. My study period was from 2001 through 2007. This means there were 84 year-month observations of each state.

4.2 Control Groups and Summary Statistics

I used three control groups: the state of Washington, a control with empirically derived weights, and an equal weight control. I estimated weights for the control group using empirically derived unit weights based on the method developed by Arkhangelsky et al. (2021), with a modification. I modified the procedure by calculating these weights using yearly means of the response variable. Without this modification, the standard estimator yielded weights that were similar to equal weighting. These weights were constructed to match Oregon's pre-2005 trend, while allowing for level differences. I then applied the unit weights to the state-level panel observations to generate the control group. I considered for inclusion in the control with empirically derived weights states in the Western Census region that did not have significant changes to their WC programs in 2001 through

[13]This dataset is a product of the Institute for Social Research and Data Innovation at the University of Minnesota. IPUMS makes analyzing the Current Population user-friendly by, for example, maintaining consistent variable coding across years.

Table 4: Pre- and Post-Treatment Characteristics of the Treatment and Control Groups

Panel A: Pre-Treatment (2001-2004)

	Oregon	Washington	Empirically Derived	Equal Weight
Unemployment Rate (%)	6.06	5.59	4.26	4.09
	(0.16)	(0.15)	(0.06)	(0.06)
Female Employment Share (%)	46.16	46.01	45.89	45.74
	(0.17)	(0.13)	(0.11)	(0.10)
Observations	48	48	384	384

Panel B: Post-Treatment (2005-2007)

	Oregon	Washington	Empirically Derived	Equal Weight
Unemployment Rate (%)	4.59	3.88	2.91	2.90
	(0.15)	(0.15)	(0.06)	(0.06)
Female Employment Share (%)	45.35	46.19	45.67	45.37
	(0.16)	(0.19)	(0.11)	(0.10)
Observations	36	36	288	288

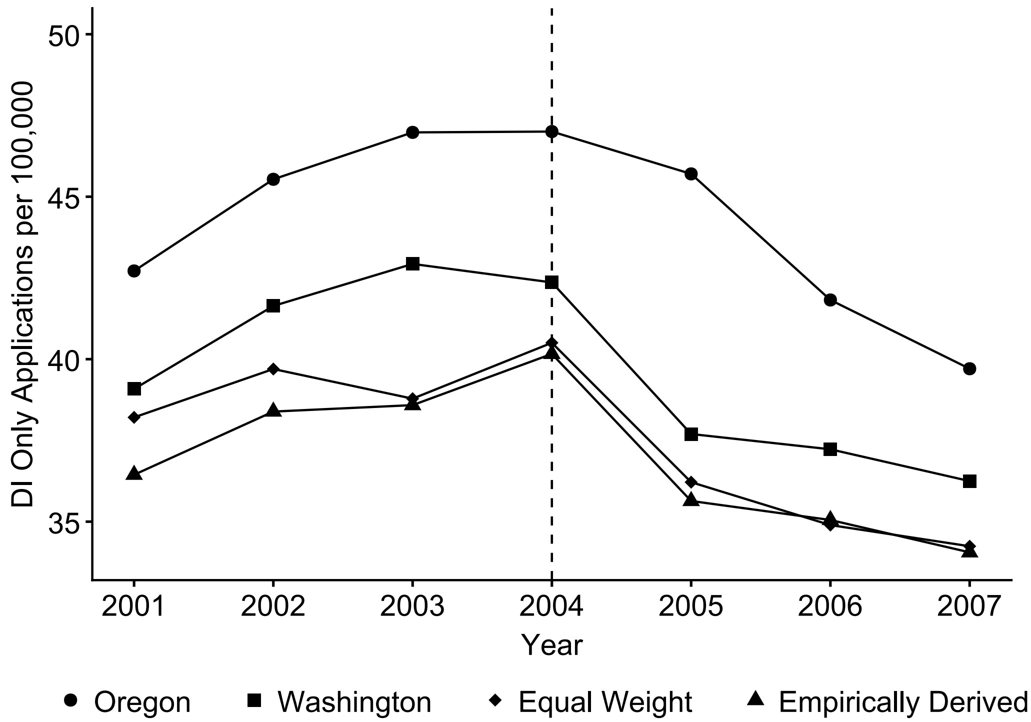
Values are sample means over the indicated time period. Standard errors are in parentheses.

2007; this resulted in eight states.[14] Table 3 shows the weight of each state that comprised the control with empirically derived weights. The control with empirically derived weights weighted Washington the most, at a weight of 0.220. As a comparison to the control with empirically derived weights, I also created an equal weight control. The equal weight control simply assigned a weight of 1/8 to each of the eight states.

To provide context on the economic environment during the study period, Table 4 presents summary statistics for the two key labor market indicators that served as covariates: the unemployment rate and the female employment share. I show pre-treatment characteristics in Panel A and post-treatment characteristics in Panel B. Columns are provided for Oregon, the state of Washington, the control with empirically derived weights, and the equal weight control. It is important to note that the empirically derived weights were constructed to minimize the difference in trends between Oregon and the eight states in the Western Census region, rather than to match pre-treatment levels.

[14] Alaska, California, Montana, and Nevada had potentially confounding changes to their WC programs. See Qiu and Grabell (2015) for specific details.

Figure 1: Observed Means Plot of Social Security Disability Insurance Only Applications



The table shows that the the unemployment rates were less in all control groups than in the treatment state of Oregon for both the pre- and post-treatment periods. On the other hand, Oregon and all control groups had similar female employment share for both the pre- and post-treatment periods.

Turning to the response variable, Figure 1 shows an observed means plot for Oregon, Washington, the control with empirically derived weights, and the equal weight control. As can be seen in the figure, Oregon and the three controls exhibited roughly parallel trends until 2005. In 2005, Oregon decreased slightly whereas Washington, the control with empirically derived weights, and the equal weight control decreased more substantially. This motivated a difference-in-differences approach because the key identifying assumption in this approach is that of parallel trends prior to treatment.

5 How did the implementation of SB 757 affect DI applications in Oregon?

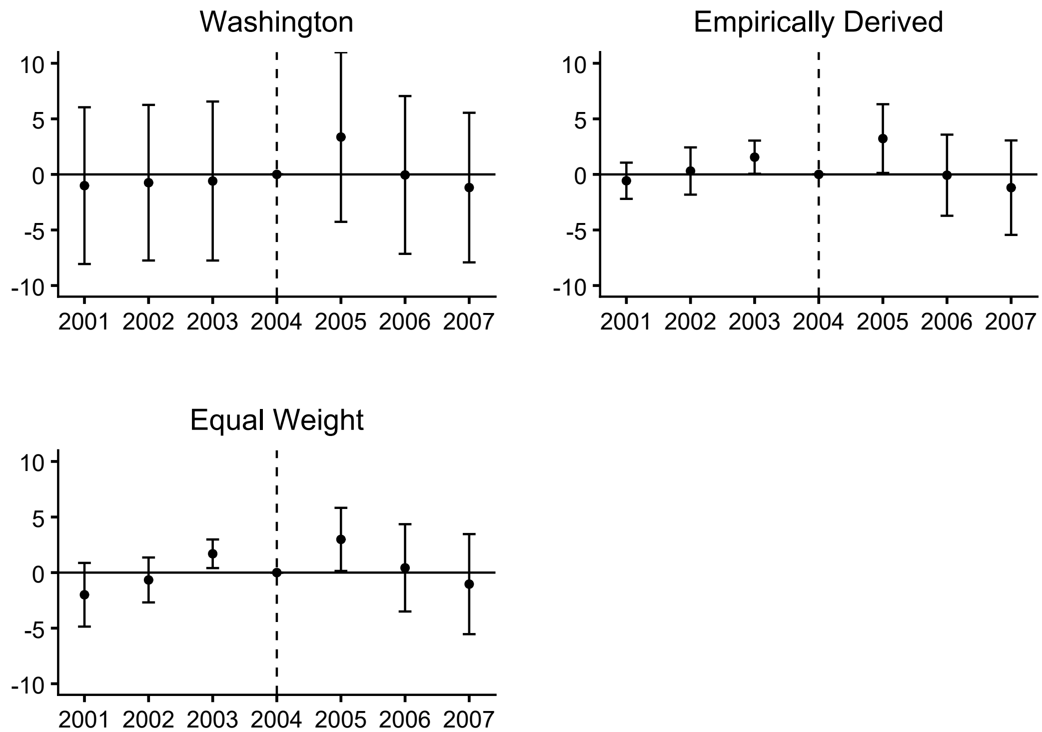
In order to more rigorously analyze the trends in the response variable, I first performed an event study against each control group separately. The event study model was the following:

$$y_{stm} = \alpha + \boldsymbol{\beta_1}(\text{Oregon}_s \times \mathbf{year}_t) + \gamma_s + \gamma_t + \gamma_m + \varepsilon_{stm} \tag{1}$$

In equation 1, subscripts s, t, and m denote state, year, and month. The treatment state is Oregon. The control group is one of the following: the state of Washington, the control with empirically derived weights, or the equal-weight control. y is the number of DI-only applications per 100,000 adults aged 25-64. α is the constant term. Oregon is an indicator for observations of Oregon. **year** is a vector of year indicators with 2004 left out to avoid perfect multicollinearity. The terms γ_s, γ_t, and γ_m are state, year, and month fixed effects. ε is idiosyncratic error.

Figure 2 plots the results of this event study. Specifically, the results were the $\boldsymbol{\beta_1}$ coefficients of the interaction term (Oregon $\times$ **year**) with their associated 95 percent confidence intervals. When the state of Washington was the control, no year was statistically significant; however, the positive sign of the point estimate of 3.36 (CI: -4.27–10.99) for 2005 was consistent with my hypothesis. Against the control with empirically derived weights and equal weight control groups, the point estimates of 3.22 (CI: 0.11–6.33) and 2.99 (CI: 0.13–5.84) were positive and statistically significant at the 5 percent level for 2005. However, the point estimates of 1.55 (CI: 0.05–3.05) and 1.69 (CI: 0.40–2.99) for 2003 against these two control groups were also positive and statistically significant at the 5 percent level. This implied that controlling for time-invariant characteristics of Oregon and the states included in the control with empirically derived weights and equal weight controls may have led to a violation of the parallel trends assumption. It followed that the estimate of the treatment effect obtained via difference-in-differences for the control with empirically derived weights and equal weight control would not represent the causal effect of treatment. Rather, the

Figure 2: Event Study Plots



estimates would capture the treatment effect and preexisting differences in trends.

To further examine how DI applications respond to changes in WC policies, I estimated the following difference-in-differences equations:

$$y_{stm} = \alpha + \beta_1\{\text{Oregon}_s \times \mathbb{1}(\text{year} \geq 2005)_t\} + \gamma_s + \gamma_t + \gamma_m + \varepsilon_{stm} \tag{2}$$

$$y_{stm} = \alpha + \beta_1\{\text{Oregon}_s \times \mathbb{1}(\text{year} \geq 2005)_t\} + \mathbf{X}_{stm}\boldsymbol{\beta_2} + \gamma_s + \gamma_t + \gamma_m + \varepsilon_{stm} \tag{3}$$

Here, $\{\text{Oregon} \times \mathbb{1}(\text{year} \geq 2005)\}$ is an interaction term that indicates observations of Oregon on or after January 2005, i.e., observations treated by the implementation of SB 757. The coefficient β_1 is the difference-in-differences coefficient and was the key coefficient of interest in the analyses. **X** was a vector of the two covariates: unemployment rate and female employment share. The rest of the terms have the same definitions as for the event study in equation 1.

Table 5 displays the results of estimating equations 2 and 3. The main coefficient of interest was

Table 5: Difference-in-Differences Estimation Results

Panel A: Estimates from Equation 2

	Washington	Empirically Derived	Equal Weight
Oregon $\times$ $\mathbb{1}$(year $\geq$ 2005)	1.30	0.33	1.03
	(1.72)	(1.33)	(1.73)
N	168	756	756

Panel B: Estimates from Equation 3

	Washington	Empirically Derived	Equal Weight
Oregon $\times$ $\mathbb{1}$(year $\geq$ 2005)	0.69	0.22	0.81
	(1.79)	(1.33)	(1.71)
Unemployment Rate	0.52	0.08	−0.46
	(0.61)	(0.28)	(0.46)
Female Employment Share	−0.49	−0.20	−0.23
	(0.44)	(0.11)	(0.13)
N	168	756	756

Response variable: DI only applications per 100,000 adults aged 25-64.
Robust standard errors in parentheses for Washington.
Clustered standard errors in parentheses for the empirically derived and equal weight controls.

that on the {Oregon $\times$ $\mathbb{1}$(year $\geq$ 2005)} variable. This coefficient indicated the additional number of DI applications per 100,000 adults aged 25-64 as a result of SB 757, assuming that the parallel trends assumption holds. For equation 2, when Washington was the control, there were an additional 1.30 applications per 100,000. When compared to the control with empirically derived weights or the equal weight control, there were an additional 0.33 or 1.03 applications per 100,000. None of these coefficients were statistically significant, however, so I could not rule out zero effect. Adding the two covariates did not affect whether the coefficient estimates were statistically significant, so again I could not rule out zero effect. Additionally, the point estimates were similar to the estimates without covariates. For equation 3, when Washington was the control, there were an additional 0.69 applications per 100,000. When compared to the control with empirically derived weights or the equal weight control, there were an additional 0.22 or 0.81 applications per 100,000.

A study by the Workers' Compensation Division of the Oregon Department of Consumer and Business Services potentially explained the reasons for these (null) findings (Department of

Consumer and Business Services, 2006). That study found that 26 percent of applications for WC benefits under the SB 757 provisions received work disability awards. However, while the average PPD award in its study sample increased, the increase was not statistically significant compared to prior law. In practice, this suggests that the additional OIIs made eligible for work disability did not substantially increase DI applications along the lines I hypothesized.

6 Conclusion

This paper examined the spillover effects from a change in state WC policy to DI application rates. Using a difference-in-differences approach, I exploited the implementation of SB 757 in Oregon, which changed the calculation of PPD benefits. My results showed a treatment effect that was statistically indistinguishable from zero. While this null result ran contrary to my hypothesis that more generous WC benefits would increase DI applications, it was consistent with a state administrative study finding that the reform did not significantly increase the average PPD award. My findings suggested that policies designed to alter WC generosity might not produce spillovers to DI if their effects were not substantial enough.

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